VALSOFT CORPORATION

AI Engineer
Technical Assessment

Candidate Take-Home Exercise | February 2026 | Estimated: 3–5 hours

Assessment Overview

You will design and build a working AI automation workflow using free, open-source tools.
The scenario is synthetic — no proprietary Valsoft data or systems are required.
We evaluate your thinking, your architecture choices, and the quality of your deliverable, not just whether the workflow runs.

1. Context & Background

At Valsoft, our AI engineers design and deploy multi-step agentic workflows that automate high-volume, decision-rich business processes. These workflows span data ingestion, AI-driven triage, structured output generation, and downstream system actions — applied across support, services, and product contexts in our 160+ portfolio companies.

This assessment simulates that environment using a realistic but fully synthetic scenario. You will not need access to our systems, our data, or our infrastructure. Everything required to complete this assessment is available via free or open-source tools.

2. The Scenario

You are building an AI-powered intake and triage pipeline for an imaginary B2B software company called "ArcVault." ArcVault receives unstructured inbound requests from customers every day — via email, web form, or support portal — covering a mix of bug reports, feature requests, billing questions, and general inquiries.

The problem: requests arrive unstructured, are manually sorted by a human, and frequently end up in the wrong queue. Your goal is to automate the intake, classification, enrichment, and routing of these requests using an agentic workflow.

2.1 Sample Inbound Requests (Synthetic)

Use the following five sample inputs as your test data. You may add more if useful.

#	Source	Raw Message
1	Email	Hi, I tried logging in this morning and keep getting a 403 error. My account is arcvault.io/user/jsmith. This started after your update last Tuesday.
2	Web Form	We'd love to see a bulk export feature for our audit logs. We're a compliance-heavy org and this would save us hours every month.
3	Support Portal	Invoice #8821 shows a charge of $1,240 but our contract rate is $980/month. Can someone look into this?
4	Email	I'm not sure if this is the right place to ask, but is there a way to set up SSO with Okta? We're evaluating switching our auth provider.
5	Web Form	Your dashboard stopped loading for us around 2pm EST. Checked our end — it's definitely on yours. Multiple users affected.

3. What You Must Build

Build a working, end-to-end agentic workflow that processes each of the five sample inputs above and produces the required outputs. The workflow must be built using free or open-source tooling only.

3.1 Required Workflow Steps

Step 1 — Ingestion

Accept each raw message as input. You may use a webhook trigger, a watched folder, a form submission, or any other trigger mechanism. The point is that the workflow starts automatically when a new message arrives.

Step 2 — Classification

Use an LLM to classify each message. At minimum, assign: (a) a Category — one of: Bug Report, Feature Request, Billing Issue, Technical Question, Incident/Outage; (b) a Priority — Low, Medium, or High; (c) a Confidence Score — how certain the model is about its classification.

Step 3 — Enrichment

Extract and structure key entities from the message. This should include: the core issue in one sentence, any relevant identifiers mentioned (account IDs, invoice numbers, error codes, etc.), and an urgency signal based on the message content.

Step 4 — Routing Decision

Based on the classification, determine the correct destination queue. Define at least three destination queues (e.g., Engineering, Billing, Product, IT/Security) and write routing logic that maps classifications to queues. Implement a fallback for low-confidence classifications.

Step 5 — Structured Output

For each message, produce a structured JSON record containing all fields from steps 2–4, plus a human-readable summary (2–3 sentences) suitable for the receiving team. Write this output to a file, a Google Sheet, a mock webhook, or any persistent destination.

Step 6 — Human Escalation Flag

If the model's confidence is below 70%, or if the message matches defined escalation criteria (e.g., the word 'outage', 'down for all users', 'billing error > $500'), flag the record for human review and route it to a separate escalation queue instead of the standard destination.

3.2 Recommended Free Tools

You may use any combination of the following (or equivalents). You are not required to use all of them.

Tool / Platform	Role in Workflow
n8n (self-hosted or cloud free tier)	Workflow orchestration, triggers, routing logic
OpenAI API / Gemini/ Groq / Ollama	LLM classification, enrichment, summary generation
Google Sheets (free)	Structured output storage, escalation queue
Webhook.site	Simulating downstream system endpoints
Gmail or Mailhog	Email trigger simulation
Airtable (free tier)	Structured record storage alternative

Zapier / Make (free tier)	Alternative orchestration if preferred

4. What You Must Deliver

4.1 Working Workflow

Provide one of the following so we can see the workflow in action:

- A screen recording (Loom or equivalent) walking through the workflow end-to-end with all five sample inputs processed
- A shareable link to your n8n workflow (exported JSON) + screenshots of each step with output
- A live demo during your technical interview (notify us in your submission if you prefer this)

4.2 Structured Output File

Submit the output records for all five sample inputs. This should be a JSON file or a shared Google Sheet showing each record with all enriched fields populated. We should be able to see: category, priority, confidence score, extracted entities, routing destination, escalation flag, and human-readable summary.

4.3 Prompt Documentation

Include the prompts you wrote for each LLM step. For each prompt, provide a one-paragraph explanation of why you structured it the way you did — what tradeoffs you made and what you would change with more time.

4.4 Architecture Write-Up

A short document (one to two pages) covering:

- Your system design: how the pieces connect, what triggers what, where state is held
- Your routing logic: how you mapped classifications to queues and why
- Your escalation logic: the criteria you chose and why
- What you would do differently at production scale (reliability, cost, latency)
- What you would add in a Phase 2 if you had another week

5. Evaluation Criteria

Your submission will be evaluated across five dimensions. Each is weighted independently — a strong architecture write-up can compensate for a rough demo, and vice versa.

Dimension	What We're Looking For	Weight
Workflow Functionality	Does the workflow run? Does it correctly process all five inputs and produce the right outputs for each step?	**25%**
Classification & Prompt Quality	Are the LLM outputs accurate and consistent? Are the prompts well-structured, with clear instructions and output formatting?	**25%**
System Design Thinking	Is the workflow well-architected? Is the routing logic coherent? Are edge cases handled (low confidence, escalation)?	**20%**
Structured Output Quality	Are the output records clean, complete, and usable? Would a downstream team be able to act on them without additional work?	**15%**
Documentation & Communication	Is the write-up clear and concise? Does it show genuine understanding of the tradeoffs — not just what was built, but why?	**15%**

What We're Really Evaluating

We are less interested in whether your workflow is perfect and more interested in how you think.

A candidate who builds a clean 4-step workflow and explains exactly why they made each decision

will score higher than one who builds all 6 steps but cannot articulate the reasoning behind them.

Show your thinking. Document your assumptions. Be honest about what you'd do differently.

6. Submission Instructions

1. Send your submission to your recruiting contact at Valsoft with the subject line: AI Engineer Assessment — [Your Name].
2. Include all deliverables from Section 4 either as attachments or as shared links (Google Drive, Notion, GitHub, Loom, etc.).

3. Submit within 2 business days of receiving this assessment. If you need more time, let us know before the deadline — we are flexible for candidates who communicate proactively.
4. If you have questions about the scenario or deliverables, email your recruiting contact. We will not answer questions about which tools to use or how to structure your prompts — those choices are part of the assessment.

7. Tips & Clarifications

On Scope

Do not over-engineer this. A clean, well-documented 4-step workflow beats a complex 10-step workflow that is hard to follow. We value clarity and intentionality over completeness.

On LLM Choice

You may use any LLM accessible via a free or low-cost API: OpenAI (free tier or pay-as-you-go), Groq (free tier), Mistral, Ollama running locally, or any equivalent. State which model you used and why in your write-up.

On the Output Destination

We do not care where the structured records land as long as we can see them. A Google Sheet, a JSON file, a Webhook.site capture, or even a well-formatted screenshot of the output node in n8n is acceptable.

On Estimated Time

This assessment is designed to take 3–5 hours. If you are spending significantly more than that, you are likely overbuilding. Timebox yourself and submit what you have with notes on what you would do next.

On Use of AI Tools

You may use AI tools (including Claude, ChatGPT, Copilot, etc.) to help you build this workflow. We expect you to. What we will ask you about in the interview is: why you made the decisions you made, and what the AI got wrong that you had to fix.

Good luck. We look forward to seeing how you think.

Valsoft AI Team | February 2026